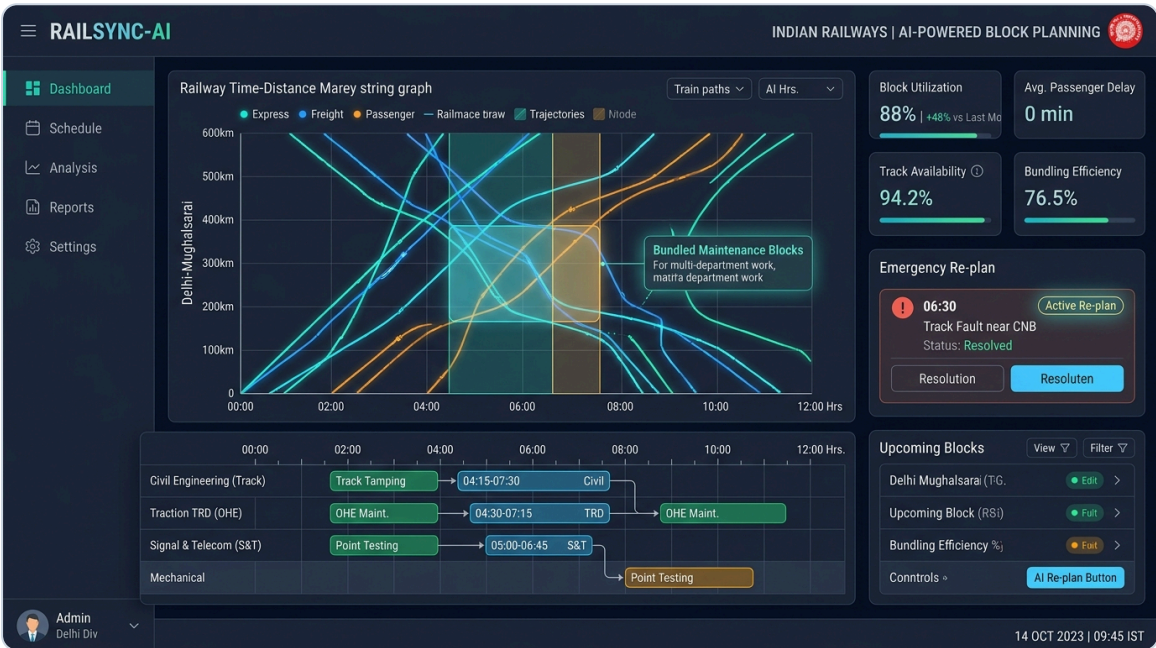


RAILSYNC-AI: System Cockpit & Dashboard Prototype

Below is the high-fidelity UI prototype designed for the **Smart India Hackathon (Ministry of Railways - SIH26027)** decision-support command center.

Controller Dashboard Mockup



RAILSYNC-AI Dashboard Cockpit

Visual Breakdown of the Interface

1. Central Viewport: Time-Distance Marey Diagram (String Graph)

- **X-Axis (Horizontal):** 24-Hour Time Horizon (00:00 to 12:00 / 24:00 Hrs).
- **Y-Axis (Vertical):** High-Density Corridor Track Distance (Delhi ↔ Mughalsarai / Kanpur Section, 0 to 600 km).
- **Train Trajectories:**
 - **Cyan Lines:** High-Priority Mail/Express Trains (*Rajdhani, Vande Bharat*) running on strict timetables.
 - **Orange Lines:** Regional Passenger services.
 - **Blue Lines:** Freight & Goods Rakes dynamically slotted via FOIS forecasting.
- **Highlighted Golden-Green Box (Bundled Maintenance Window):**
 - Represents an **Integrated Shadow Corridor Block** scheduled during the early morning low-traffic window (04:15 – 07:30).
 - Demonstrates zero intersection/delays for express passenger paths.

2. 🕒 Bottom Viewport: Multi-Department Synchronized Gantt Lane

- **Civil Engineering Lane (TMS):** Track Tamping Machine (CSM) operation scheduled from 04:15 to 07:30.
 - **Traction TRD Lane (TDMS):** 25 kV OHE power isolation and wire maintenance executed simultaneously (04:30 to 07:15) within the same civil shutdown.
 - **Signal & Telecom Lane (SMMS):** Point machine testing and track circuit verification slotted concurrently (05:00 to 06:45).
 - **Result: 3 departments completing work in 1 combined line shutdown**, cutting total track closure time by **>50%**.
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3. 📊 Top-Right Panel: Real-Time Operational KPI Cards

- **88% Block Utilization (+48% vs Last Month):** Drastic reduction in idle/unproductive block hours.
 - **0 min Avg. Passenger Delay:** 100% punctuality preservation for scheduled services.
 - **94.2% Track Availability:** Maximized line capacity for goods movement and revenue generation.
 - **76.5% Bundling Efficiency:** High proportion of tasks executed as multi-department shadow blocks.
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4. 🚨 Middle-Right Panel: Emergency Disruption & Re-Plan Module

- **Live Incident Alert:** Shows a real-time event (e.g., *06:30 Track Fault near CNB*).
- **Automated Resolution:** Google OR-Tools CP-SAT solver recalibrates and re-routes affected freight paths in **sub-second runtime**, with a 1-click **"Execute Resolution"** workflow for the Section Controller.